

Assignment No 15

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Domain : AI & ML

College : T.B Girwalkar Polytechnic Ambajogai

Aim : To read, load, explore, and understand the Countries dataset using the Pandas library in Python.

1. Import the Pandas library.

```
import numpy as np
import pandas as pd
```

```
> from google.colab import files
  uploaded = files.upload()

for fn in uploaded.keys():
    print(f'user uploaded file "{fn}" with length {len(uploaded[fn])} bytes')
```

```
** Choose Files countries.csv
countries.csv(text/csv) - 3393 bytes, last modified: 7/4/2026 - 100% done
Saving countries.csv to countries.csv
user uploaded file "countries.csv" with length 3393 bytes
```

```
filepath='/content/countries.csv'
df=pd.read_csv(filepath)
```

2. Load the CSV dataset into a DataFrame.

...

	Country	Region
0	Algeria	AFRICA
1	Angola	AFRICA
2	Benin	AFRICA
3	Botswana	AFRICA
4	Burkina	AFRICA
...	...	...
189	Paraguay	SOUTH AMERICA
190	Peru	SOUTH AMERICA
191	Suriname	SOUTH AMERICA
192	Uruguay	SOUTH AMERICA
193	Venezuela	SOUTH AMERICA

194 rows × 2 columns



3. Display the first 5 rows of the dataset.

 `df.tail()`

...

Country

Region



189	Paraguay	SOUTH AMERICA
-----	----------	---------------

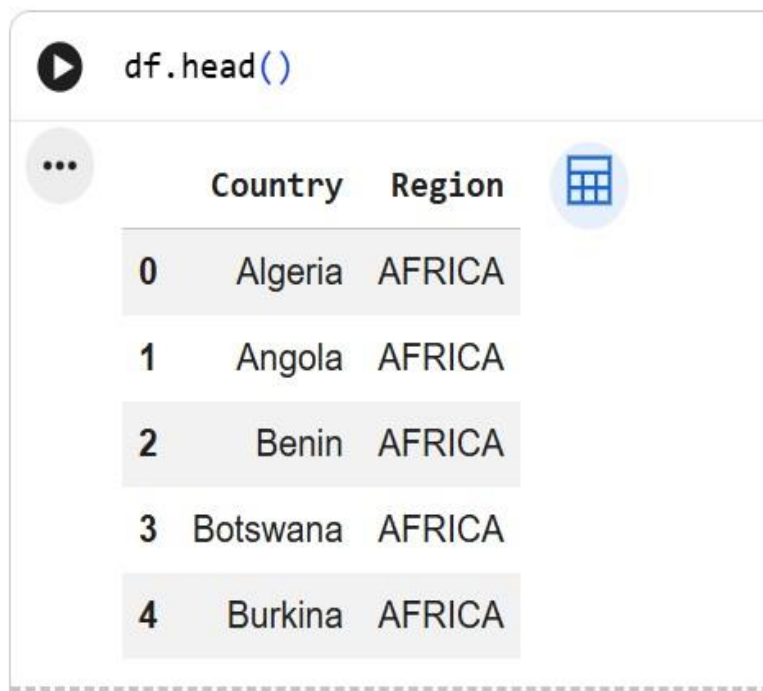
190	Peru	SOUTH AMERICA
-----	------	---------------

191	Suriname	SOUTH AMERICA
-----	----------	---------------

192	Uruguay	SOUTH AMERICA
-----	---------	---------------

193	Venezuela	SOUTH AMERICA
-----	-----------	---------------

4. Display the last 5 rows of the dataset.



The image shows a Jupyter Notebook interface. At the top, there is a code cell with the text `df.head()` and a play button icon. Below the code cell, there is a table with 3 columns: an index column, a 'Country' column, and a 'Region' column. The table contains 5 rows of data. To the right of the table, there is a blue icon of a grid. Below the table, there is a dashed line.

	Country	Region
0	Algeria	AFRICA
1	Angola	AFRICA
2	Benin	AFRICA
3	Botswana	AFRICA
4	Burkina	AFRICA

5. Show the number of rows and columns.

[19]

✓ 0s



`df.shape`

`(194, 2)`

6. Display all column names.

[21]

✓ 0s

```
df.columns
```

▼

```
Index(['Country', 'Region'], dtype='object')
```

7. Show dataset information, data types, and non-null values.

[22]

✓ 0s



df.info()



... <class 'pandas.core.frame.DataFrame'>

RangeIndex: 194 entries, 0 to 193

Data columns (total 2 columns):

#	Column	Non-Null Count	Dtype
0	Country	194 non-null	object
1	Region	194 non-null	object

dtypes: object(2)

memory usage: 3.2+ KB

8. Generate summary statistics for numerical columns.

[23]

✓ 0s



```
df.describe()
```



...

Country Region



count	194	194
-------	-----	-----

unique	194	6
--------	-----	---

top	Algeria	AFRICA
-----	---------	--------

freq	1	54
------	---	----

9. Display the data type of each column.

[24]

✓ 0s



```
df.dtypes
```

```
0
```

```
Country    object
```

```
Region     object
```

```
dtype: object
```

10. Count missing values in each column.

 `df.isnull().sum()`

... 0

Country 0

Region 0

dtype: int64

11. Count unique values in each column.

[26]

✓ 0s



```
df.nunique()
```



...

0

Country 194

Region 6

dtype: int64

12. Display 5 random rows from the dataset.

[27]

✓ 0s

```
df.sample(5)
```

CODE

TEXT

✓

	Country	Region
173	Nauru	OCEANIA
166	Trinidad and Tobago	NORTH AMERICA
185	Chile	SOUTH AMERICA
82	Pakistan	ASIA
77	Malaysia	ASIA



13. Count occurrences of unique values (for a selected column).

```
df.value_counts()
```

...		count
Country	Region	
Afghanistan	ASIA	1
Albania	EUROPE	1
Algeria	AFRICA	1
Andorra	EUROPE	1
Angola	AFRICA	1
...	...	...
Venezuela	SOUTH AMERICA	1
Vietnam	ASIA	1
Yemen	ASIA	1
Zambia	AFRICA	1

